



Examining Age-Related Differences in Neural Responses to Trust

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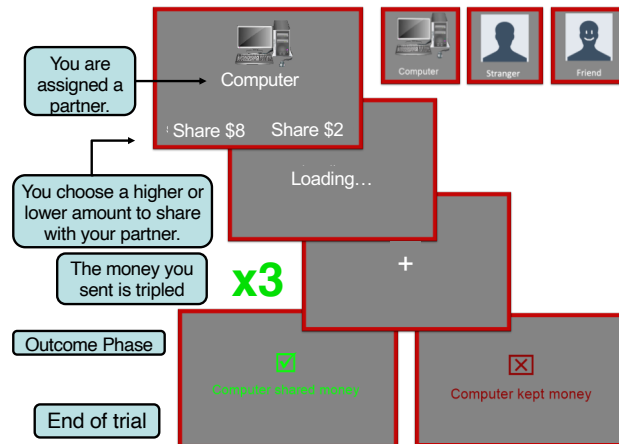
1. Introduction

- Social relationships influence our behaviors and modulate our neural responses to reward (Fareri et al., 2015).
- The effect of relationships on behavior and reward activation change as we age (Fareri et al., 2022; Cassidy et al., 2020).
- It remains unclear whether age-related differences in responses to trust are associated with maladaptive outcomes such as financial exploitation.

We investigated whether social trust decisions change as we age, and if those changes could be related to financial exploitation.

2. Methods

Participants (N = 62) completed a trust task with three partners: a computer, a stranger, and their chosen friend.



We used the Inclusion of Other in Self Scale (IOS) and the Older Adults Financial Exploitation Measure (OAFEM) to measure closeness ratings and risk for financial exploitation.

Whole brain results were thresholded at $Z > 3.1$ and Family Wise Error corrected at $p > .05$

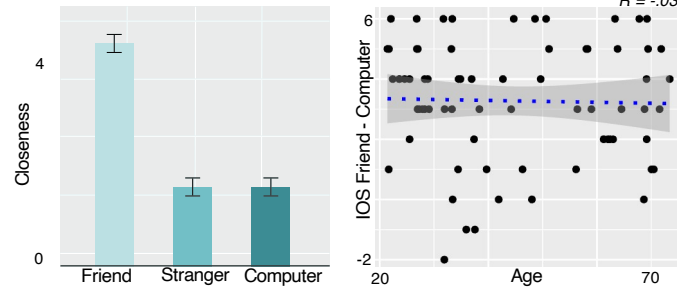
fMRI data were preprocessed with fMRIPrep and statistical analyses were performed using FSL



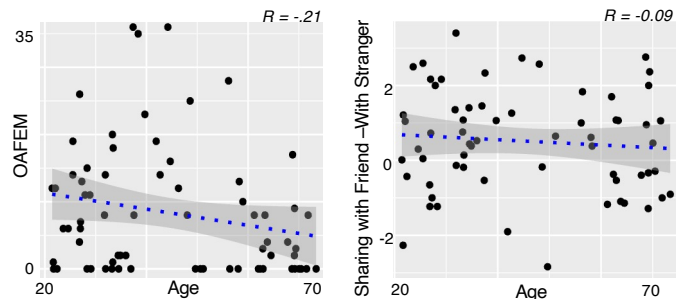
Scan the QR code to see the Pre-registration!

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3. Age-Related Behavioral Effects?

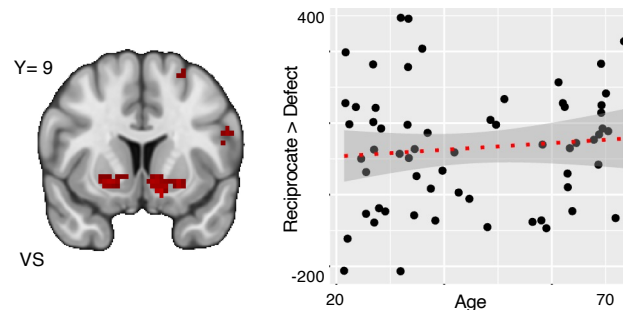


Participants are closer with their friend than with the stranger and computer partners, however, there is no relationship between age and closeness ratings.



We found no age-related effects across financial exploitation measures or investment behavior

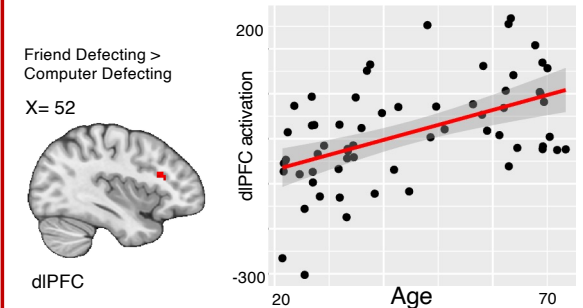
4. Age-Related Differences in Striatal Response?



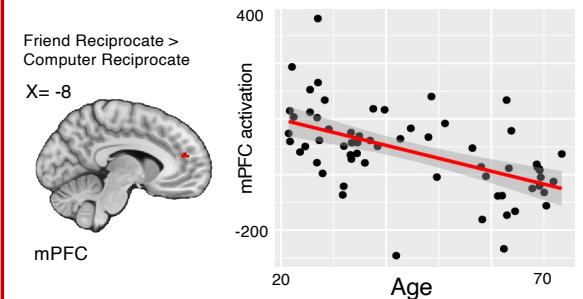
We found activation in the ventral striatum (VS) when participants were presented with a reciprocate outcome as opposed to a defect outcome, however, there was no significant association with age for any condition (Friend > Stranger and Friend > Computer).

5. Whole Brain Analyses of Neural Responses to Social Outcomes

To investigate if age-related effects were present elsewhere, we conducted a set of whole brain analyses.



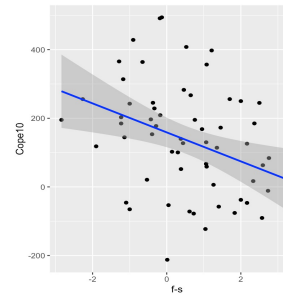
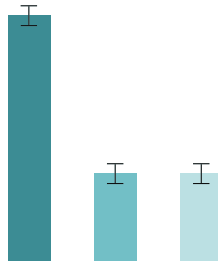
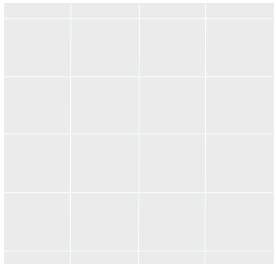
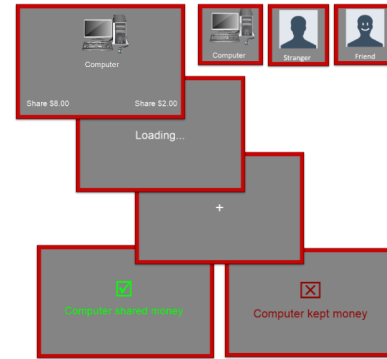
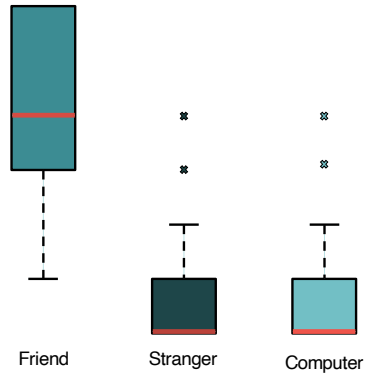
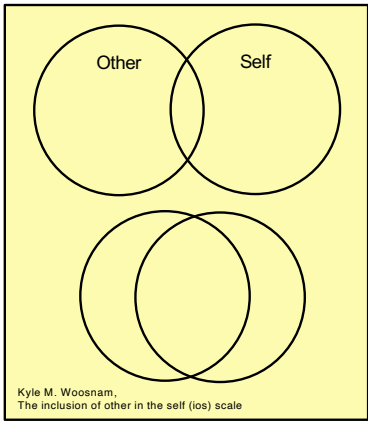
We found that friend defecting resulted in increased dorsolateral prefrontal cortex (dIPFC) activation in older adults compared to younger adults.



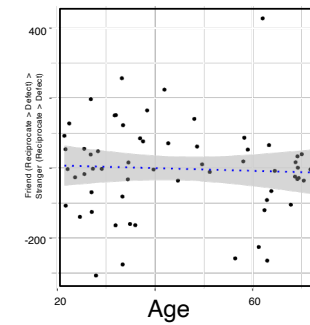
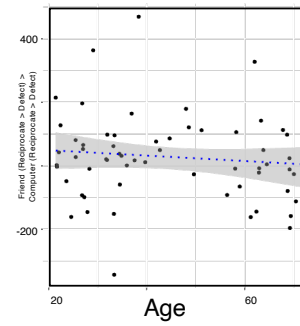
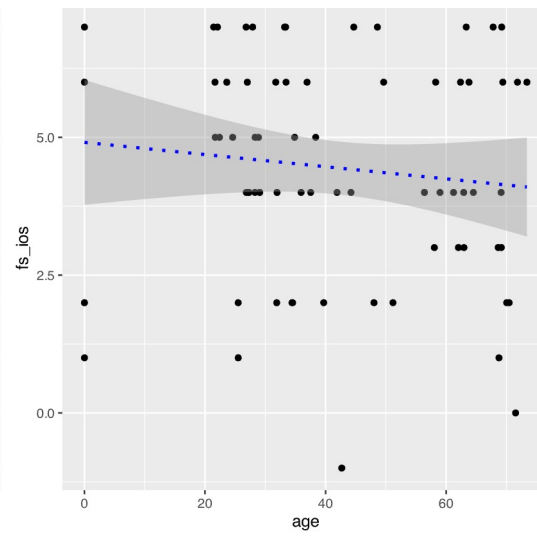
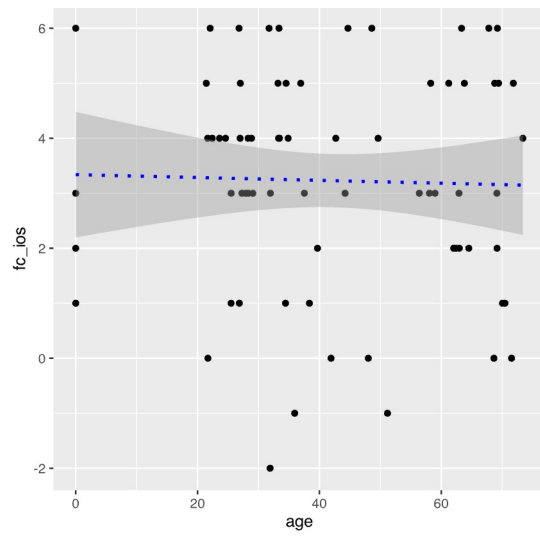
We also found that friend reciprocating resulted in reduced medial prefrontal cortex (mPFC) activation in older adults compared to younger adults.

6. Conclusions and Future Directions

- Our preliminary results indicate that **reciprocation elicited activation in the ventral striatum**.
- Additionally, we find that activation of the **dIPFC increased**, and **mPFC decreased**, during the processing of social outcomes **as a function of age**.
- Future analyses will examine how neural and behavioral responses to trust are associated with risk for financial exploitation.



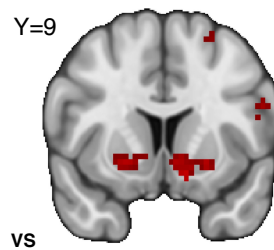
VS mask Cope 10, f-s behavior



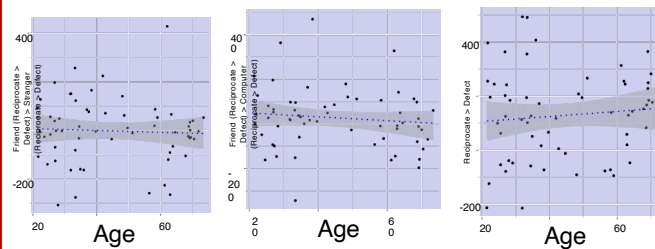
Participants were asked to rate their closeness with the three partners before the task on a 7-point Likert Scale

4. Ventral Striatal Activation Is not Associated with Age

We found Ventral Striatum (VS) activation for Reciprocate > Defect outcomes



We found activation in the VS when participants were presented with a reciprocate outcome as opposed to a defect outcome.



We found that activation in the VS was not related to age